(Please write your Enrollment Number) Enrollment No. 11020117 2022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
B.Tech :<>, Semester:<7>
(December, 2024)

Subject Code: BAI-301	Subject: Machine Learning
Time : 3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mapping
	a) Why is data cleaning an essential step in the machine learning pipeline?		1
	b) What is the importance of evaluation metrics in machine learning?		1
	c) What is the difference between Ridge Regression and LASSO Regression?		2
	d) Define the kernel trick in Support Vector Machines.		2
	e) What is k-means clustering, and how is it different from hierarchical clustering?		3
	f) Define Dimensionality Reduction.		3
	g) What is meant by cross-validation, and why is it important?		4
	h) Define ensemble learning and list its types.		4
	UNIT I		CO Mapping
Q2	Explain the Machine Learning Cycle with an example, highlighting key stages like Train-Test Split, Validation Data, and K-Fold Cross Validation.	(10)	1
Q3	Discuss the process of Data Exploration and Pre-processing, including Data Objects and Attributes, Visualization, and Feature Extraction.	(10)	1
	UNIT II		CO Mapping
Q4	Explain Logistic Regression in detail. How is it used for binary classification?	(10)	2
Q5	Explain Naïve Bayes Classifiers with examples.	(10)	2
	UNIT III		CO Mapping
Q6	Describe the process of Principal component analysis (PCA) and Linear Discriminant Analysis (LDA). Give examples	(10)	3
Q7	Explain the Expectation-Maximization (EM) algorithm and its application in clustering.	(10)	3
	UNIT IV		CO Mapping
Q8	Discuss Bagging, Boosting, and Stacking with appropriate examples.	(10)	4
Q9	Explain the concept of Random Forests using an example. How does it improve prediction accuracy compared to a single decision tree?	(10)	4